Chapter 4: Interest Rates – Master Summary

1. The Core Narrative

This chapter establishes the mathematical framework for valuing cash flows. The progression of logic is:

1. **Define the inputs:** Interest rates come in various forms (Treasury, Repo, LIBOR/SOFR) and must be standardized using **continuous compounding**.
2. **Build the foundation:** We cannot use raw bond yields for pricing; we must extract the theoretical **Zero-Coupon (Spot) Curve** using **Bootstrapping**.
3. **Project the future:** From the spot curve, we derive **Forward Rates**, which imply where the market prices future money.
4. **Manage Risk:** We use **Duration** (linear) and **Convexity** (curvature) to measure and hedge sensitivity to rate changes.

2. Key Theoretical Concepts

A. Continuous Compounding

This is the standard for derivative pricing because it simplifies calculus (limits, derivatives).

- **The Rule:** As compounding frequency $m \to \infty$, the growth factor becomes e^{Rt}.
- **Conversion:** To convert a discrete rate R_m to continuous R_c:

$$R_c = m \ln(1 + R_m/m)$$

B. The Zero Curve (Spot Curve)

The "Zero Rate" is the single most important number. It is the rate of interest earned on an investment that starts today and has *no intermediate payments*.

- **Why it matters:** You cannot discount a 5-year coupon bond using a single "5-year yield." You must discount the Year 1 coupon at the 1-year zero rate, the Year 2 coupon at the 2-year zero rate, etc.

C. Bootstrapping

The algorithm used to construct the Zero Curve from market data.

- **Process:** Solve for short-term rates first (using T-bills). Use those known rates to discount the early coupons of longer-term bonds, isolating the final principal payment to solve for the long-term zero rate.

D. Forward Rates

The rate implied by the current term structure for a future time period.

- **Arbitrage Logic:** If you can borrow for 2 years at 4%, or borrow for 1 year at 3% and roll it over, the "rollover" rate must make the two costs identical.
- **Formula:** $R_F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$

E. Duration & Convexity

- **Duration:** The first derivative of price with respect to yield. Good for small shifts.
- **Convexity:** The second derivative. Necessary for large shifts because the price-yield relationship is curved, not straight.
  - *Quant Note:* Positive convexity is valuable. It means your bond loses *less* value than expected when rates rise and gains *more* than expected when rates fall.

3. Cheat Sheet: Critical Formulas

Concept	Formula	Notes
Future Value (Continuous)	Ae^{Rt}	Grow an asset forward.
Present Value (Continuous)	Ae^{-Rt}	Discount a future cash flow.
Conversion (Discrete $\rightarrow$ Cont)	$R_c = m \ln(1 + \frac{R_m}{m})$	Essential for standardizing inputs.
Forward Rate	$\frac{R_2T_2 - R_1T_1}{T_2 - T_1}$	The "marginal" rate for the extra time.
FRA Value (Receiver)	$L(R_K - R_F)(T_2 - T_1)e^{-R_2T_2}$	Valuing an off-market forward contract.
Duration Price Impact	$\Delta B \approx -DB\Delta y$	Linear approximation of risk.
Convexity Adjustment	$+\frac{1}{2}CB(\Delta y)^2$	Curvature correction term.

4. Expert Context: Modern Quant Reality

This section adds current industry knowledge not fully detailed in older textbook editions.

The Death of LIBOR & Rise of SOFR

While the text likely references LIBOR, modern markets (post-2021) have largely transitioned to **SOFR** (Secured Overnight Financing Rate) in the US and **SONIA** in the UK.

- **Key Difference:** LIBOR contained bank credit risk (AA rated). SOFR is a "secured" repo rate, making it essentially risk-free. This fundamentally changes how we model "risk-free" proxies.

OIS Discounting (The "Real" Risk-Free Rate)

Historically, traders used LIBOR to discount cash flows. Today, the gold standard is **OIS (Overnight Index Swap)** discounting.

- **Why?** Collateralized derivative portfolios (like those under CSA agreements) earn the overnight rate (Fed Funds/SOFR) on posted collateral. Therefore, the true discount rate for these derivatives is the overnight rate, not LIBOR.

Multi-Curve Framework

In the past, we had one yield curve. Today, quants must manage **Multiple Curves**:

1. **Projection Curve:** Used to estimate future floating rates (e.g., 3-month SOFR).
2. **Discounting Curve:** Used to discount those cash flows back to today (typically OIS).

- *Consequence:* A single "Zero Curve" is no longer enough. You need a specific curve for every tenor (1-month, 3-month, 6-month) because the spreads between them are volatile.

Negative Interest Rates

The standard Lognormal models (like Black-Scholes) break if interest rates are negative (you can't take the log of a negative number). The prolonged period of negative rates in Europe/Japan forced quants to shift to **Normal (Bachelier)** or **Shifted-Lognormal** models to handle negative forward rates.